



Transforming Lifelong Learning through Precision Education

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September 18, 2026

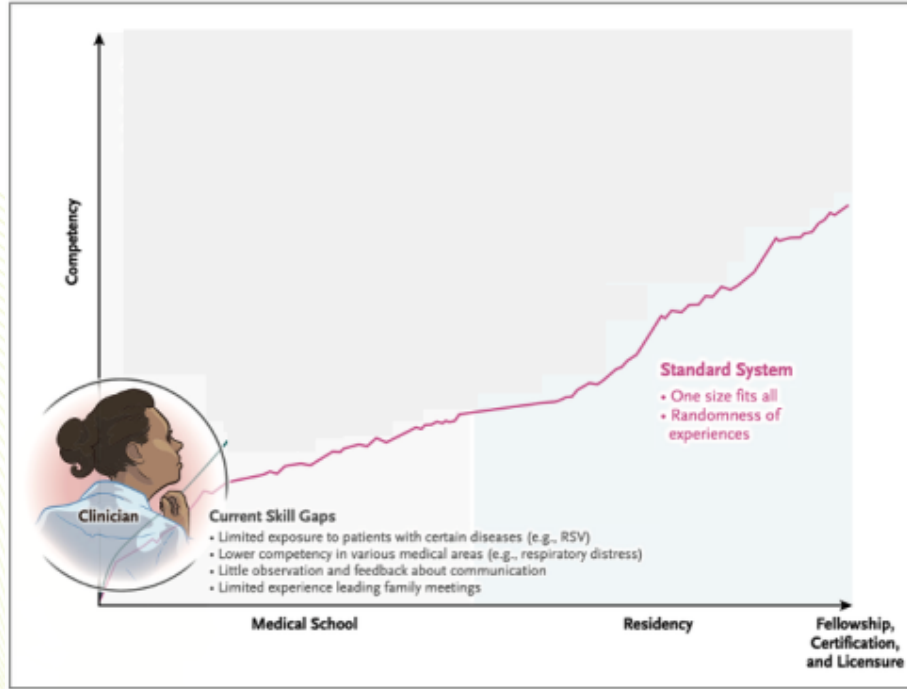


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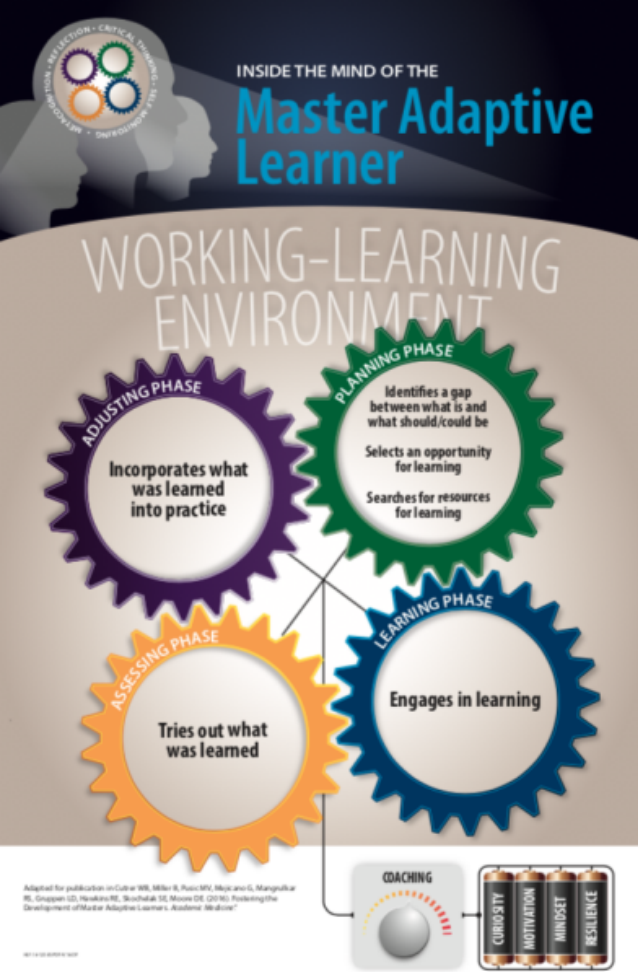
Disclosures

- I am a paid employee of the American Medical Association
- I was not observed as a trainee



The typical physician's path to excellence is fragmented

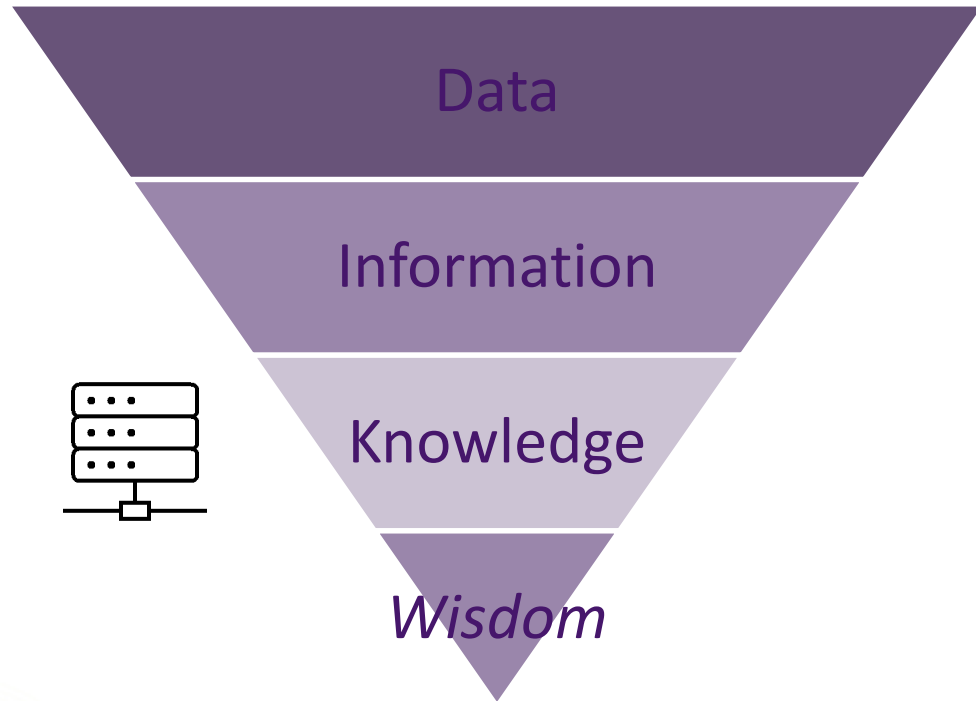
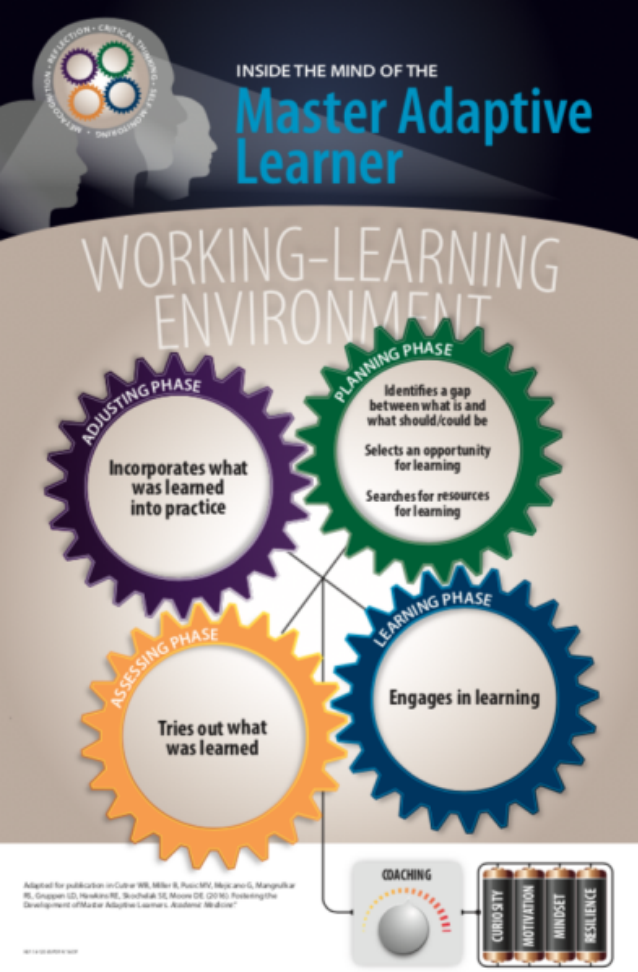
Desai, Khan, Lomis. NEJM 394;9 February 26, 2026



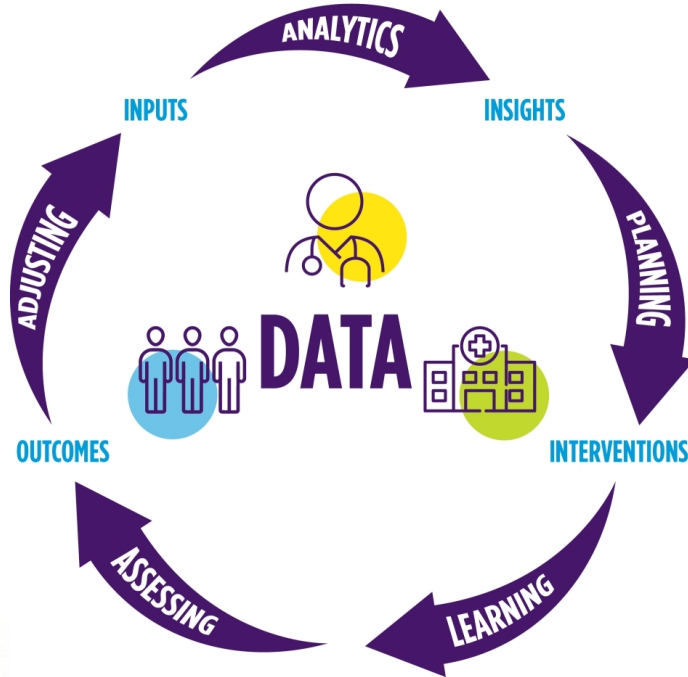
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Precision Education Framework



Precision Education Systems

emphasize the contribution of data & analytics to propel the entire learning cycle

Levels of engagement:

- individual (micro)
- program (meso)
- organization (macro)



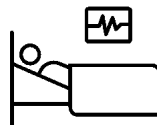
Precision education systems leverage data and technology, including AI, to make individual learning more relevant and effective across the continuum of health professions training.

We developed a \$12M four-year precision education grant program with three goals

Develop and implement precision education systems and democratize those systems across the continuum of medical education by:



Catalyzing development of **tech-enabled assessments** to inform precision education

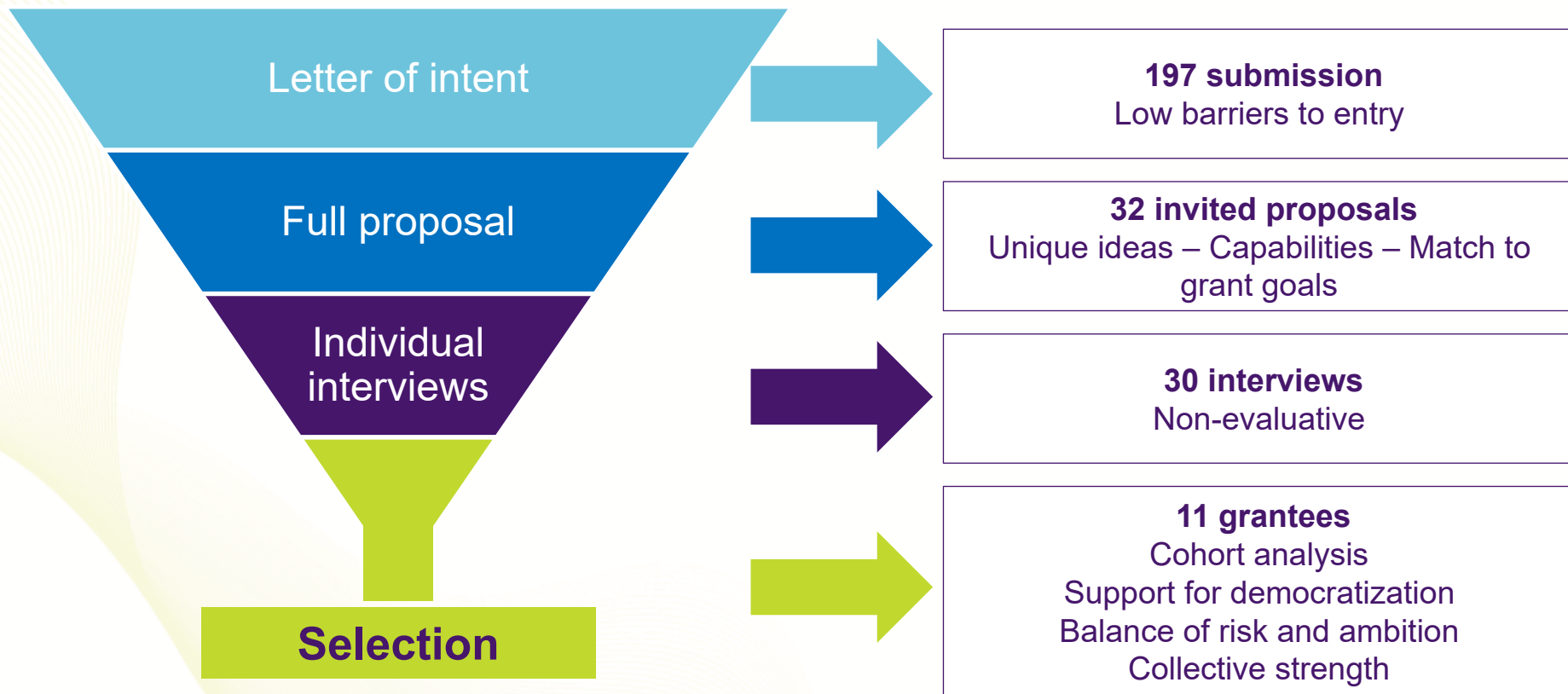


Elevating the **competencies that matter** most in serving patients and communities



Developing actions from the assessment data, **personalizing training experiences** to nurture desired competencies

We emphasized different elements at different points in the process



The 11 selected projects fund a large and diverse set of grantees

Grantee	Goal	# Partners
 Georgia Academy of Family Physicians	Individualized Learning Plans (precision education platform for family medicine residents in 12 programs across the entire state)	13
 Louisiana State University Health Sciences Center	Compassion in Motion: Training Tailored Communication Across the Clinical Continuum (builds on virtual simulation technology)	2
 Meritus School of Osteopathic Medicine	Opening the Doors with the Right Education for the Right Student at the Right Time: The Continuum of Precision Education in Maryland's Newest Medical School (a platform for brand new osteopathic school in the state)	-
 Mount Sinai	Engaging Patients Toward Better Health: Using AI-based Ambient Listening to Improve Communication Skills of Trainees and Faculty in an Underserved Community-Based Setting	-
 Stanford University	Implementation and Evaluation of a Scalable, Sensor-Enabled Precision Learning Curriculum for Procedural Skills Certification Across the Continuum (uses wearable technology and engages boards)	2 ABMS, ABS
 University of Cincinnati	Ambient AI for Precision Feedback: Augmenting Clinical Reasoning and Communication using Real-Time Feedback	2 Inc. Arizona State SoM & Engin

The 11 selected projects fund a large and diverse set of grantees

Grantee	Title	# Partners
 University of Hawaii	Culturally Responsive AI for Rural Health: Utilizing Precision Education and Precision Coaching to Strengthen Hawaii's Rural Physician Workforce (engages indigenous population and rural setting)	-
 University of Illinois	Enhancing Precision Education Using "Feed-Back" and "Feed-Forward" Learning Analytics Data Systems: A Multisite Consortium Project to Align UME-GME Transitions (data sharing bw UME GME; IMGs included)	14 Inc. ACGME, Intealth
 University of Michigan	PACE: Precision Analytics for Coaching Effectiveness (data analytics platform to drive clinical coaching in anesthesia residencies nationally)	36
 University of Pennsylvania	AI-Driven, Context-Aware, Precision Assessment of Individual and Team-Based Clinical Reasoning (uses ambient AI)	1
 University of Wisconsin	Graduate Performance Profiles to Enhance Learning Analytics and Precision Program Improvement in Vascular Surgery (links GME assessment national early practice performance data)	7 Inc. ACGME

Portfolio addresses three grant goals (continuum mapped initial focus → extension)

Over 80 institutions are engaged across the 11 grants

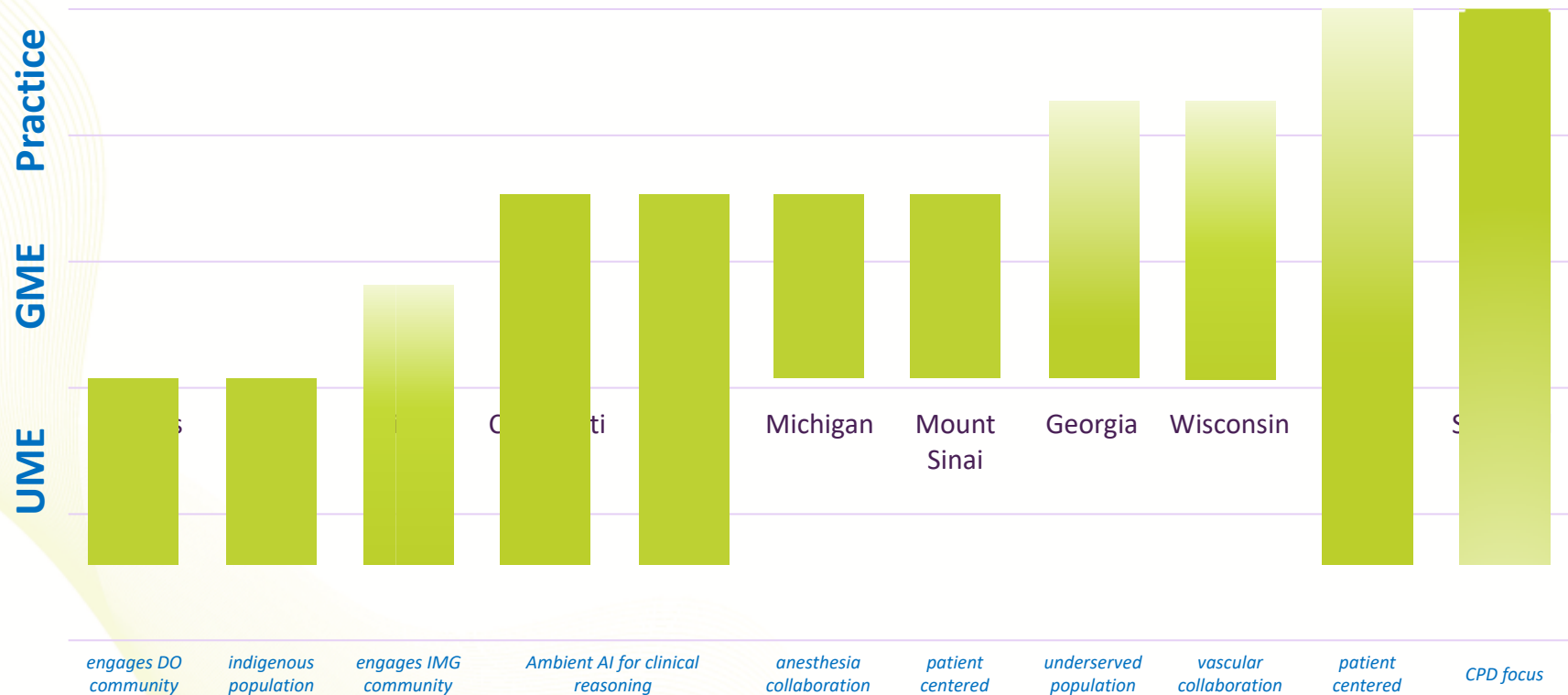
Grant Goals

Catalyze the development of tech-enabled assessments	 multi	→	 multi	→	 multi
Nurture competency in one or more domain(s) beyond medical knowledge	 →		 →		
Develop processes to act upon assessment data, personalizing educational experiences to support desired learning outcomes		 multi	→	 multi	 multi

Medical students Transition to residency Residents Transition to practice Practicing physicians

*this is a simplified representation of goals;
all projects are multi-dimensional

The projects span the full continuum from UME to Practice...



Grantees are using several different technical solutions

Dashboards

Data synthesis for
performance benchmarking



New med school



UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

UME to GME across orgs



GME specialty, state



GME specialty, national



School of Medicine
and Public Health
UNIVERSITY OF WISCONSIN-MADISON

GME to practice
program improvement

Simulation



On-demand
virtual patients for
communications



Tech-enhanced
procedural
training

Predictive

Analytics

for rural practice



Ambient AI

Clinical Reasoning

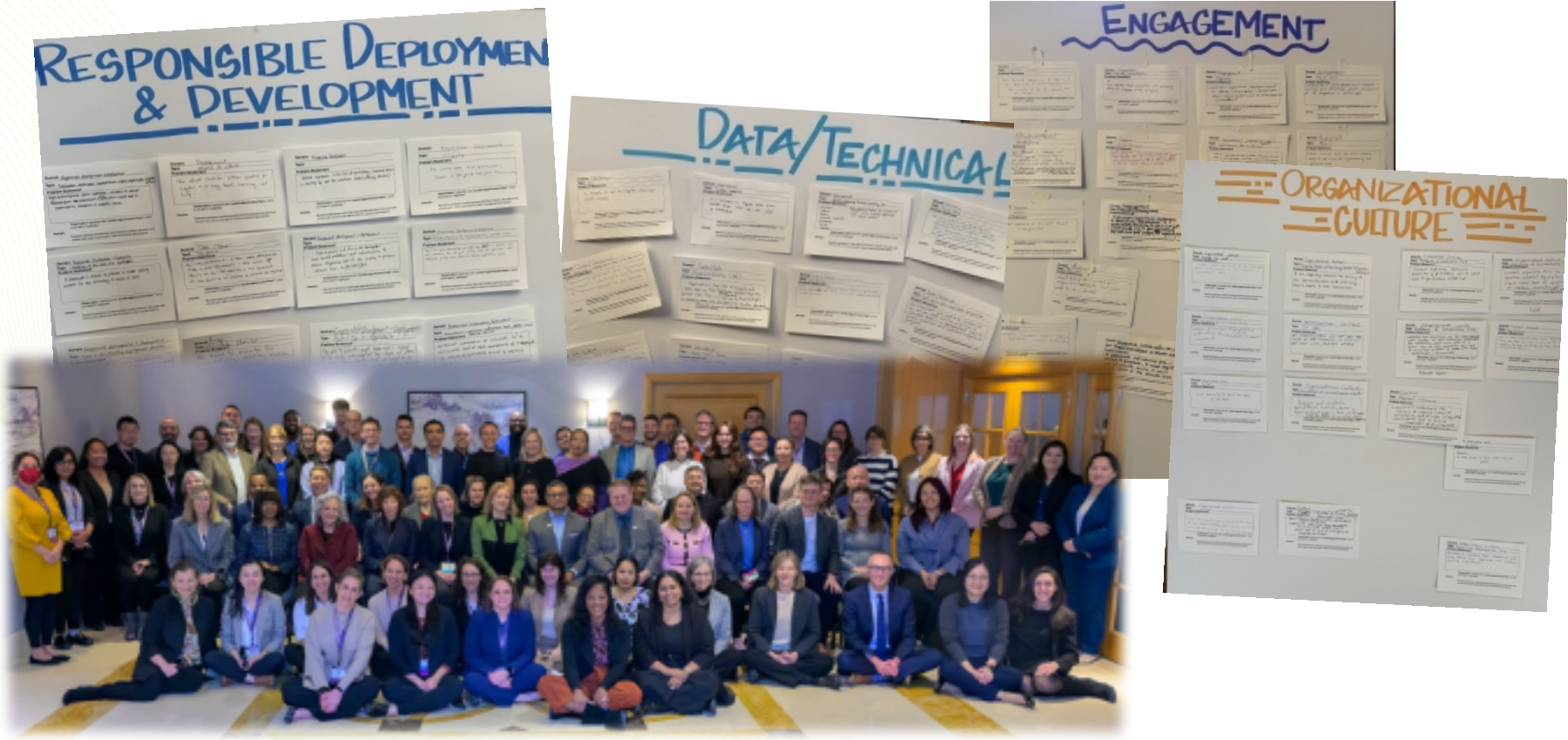


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School of Medicine
UNIVERSITY OF PENNSYLVANIA



Communications

Grantees collaborating to tackle big questions

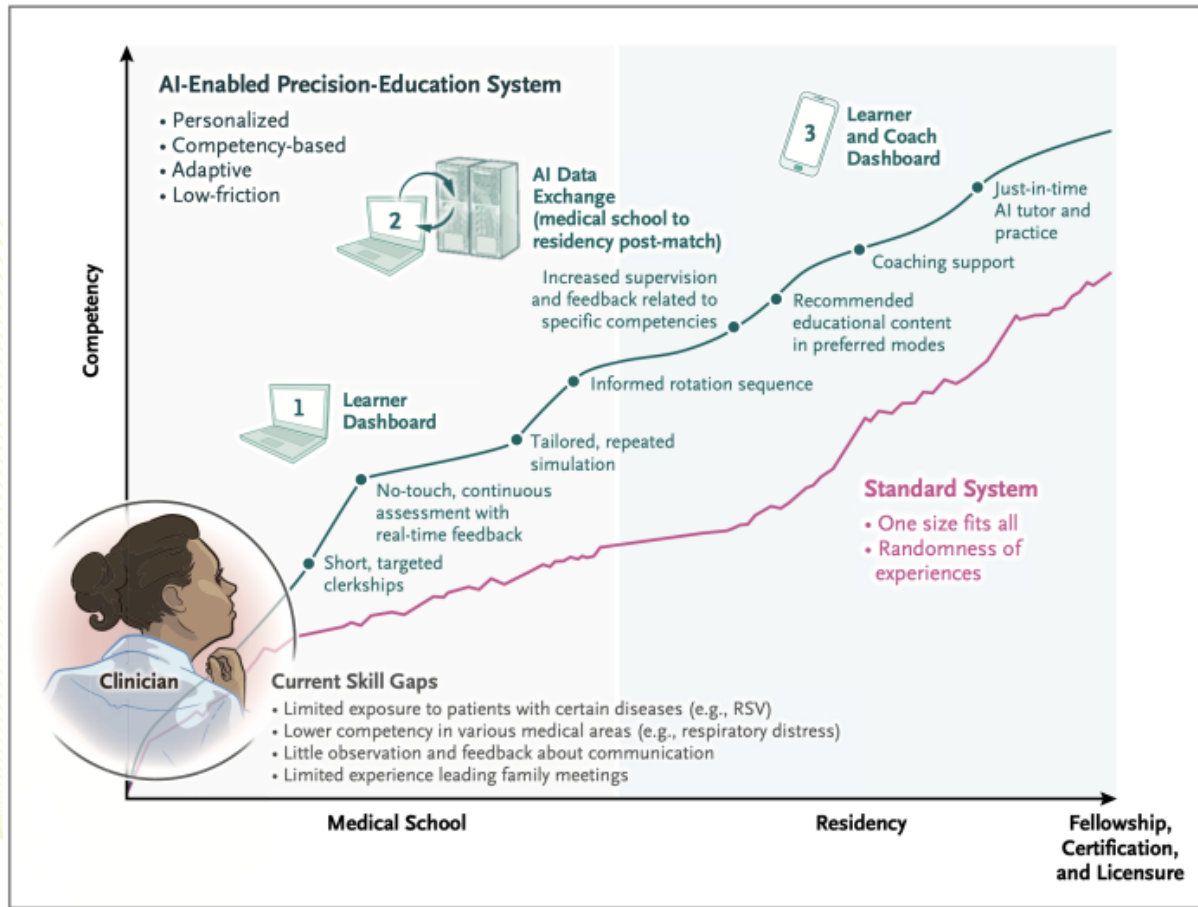


The Precision Education Consortium

What is the AMA hoping to accomplish with the program across grantees?

Set the right tone for precision education in the broader Med Ed community:

- Developmental experience across the continuum
- Learner agency: Do this *with* learners, not *to* them
- Requires innovation in culture as much as technology
- Responsible handling of data and use of AI
- Transparency about challenges and collective strategizing to mitigate them
- Democratization



Example of a Trainee's Journey under an AI-Enabled Precision-Education System.

Precision Education

leverages data & technology to enhance personalization and relevance of training across a physician's career



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**TRANSFORMING LIFELONG LEARNING
THROUGH PRECISION EDUCATION**